Network Traffic Analysis Report

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Modulo: W19D1 (Pratica)
Piattaforma: Epicode
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1. Introduction

This report presents the analysis of a provided network capture file (.pcap), with the objective of identifying Indicators of Compromise (IOCs), determining potential attack vectors, and recommending mitigation actions.

2. Traffic Analysis

The network traffic was analyzed using Wireshark. The following patterns were observed:
- High volume of TCP SYN packets
- Frequent RST, ACK responses
- Repetitive communication between IP addresses 192.168.200.100 and 192.168.200.150
- Connection attempts across multiple destination ports

These patterns indicate abnormal and non-standard network behavior.

Figure 1 – TCP SYN packets observed during scanning activity

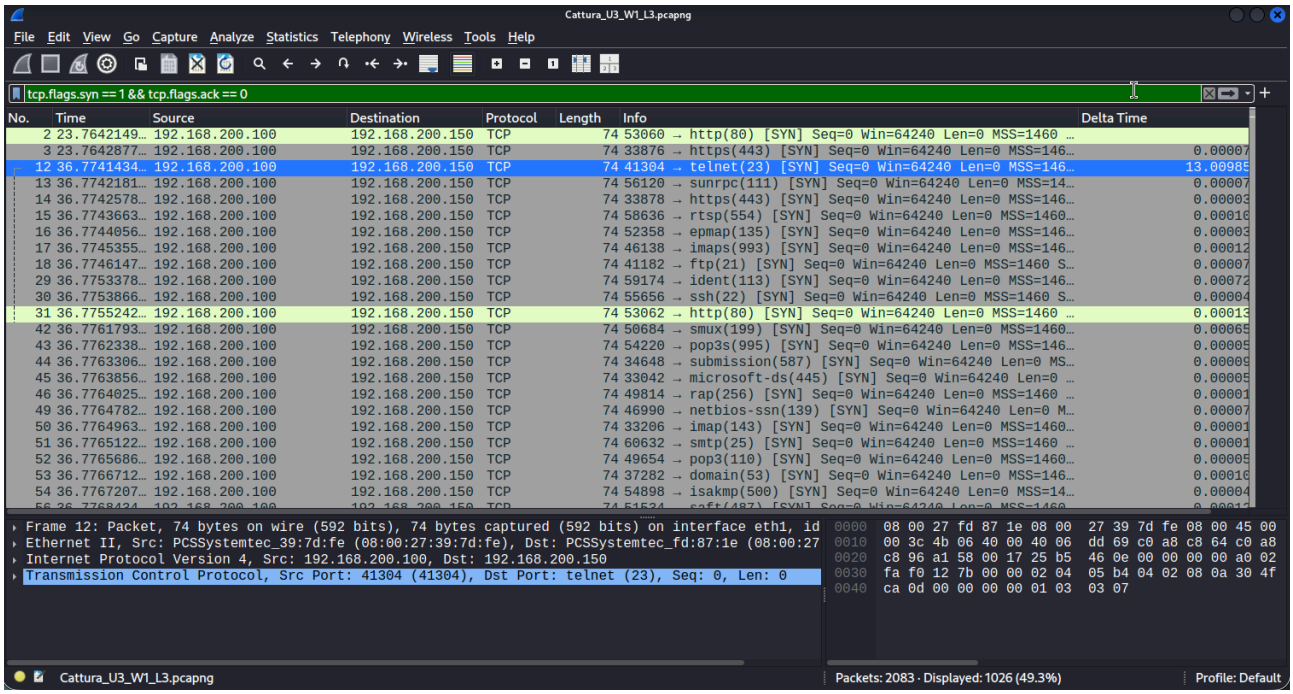
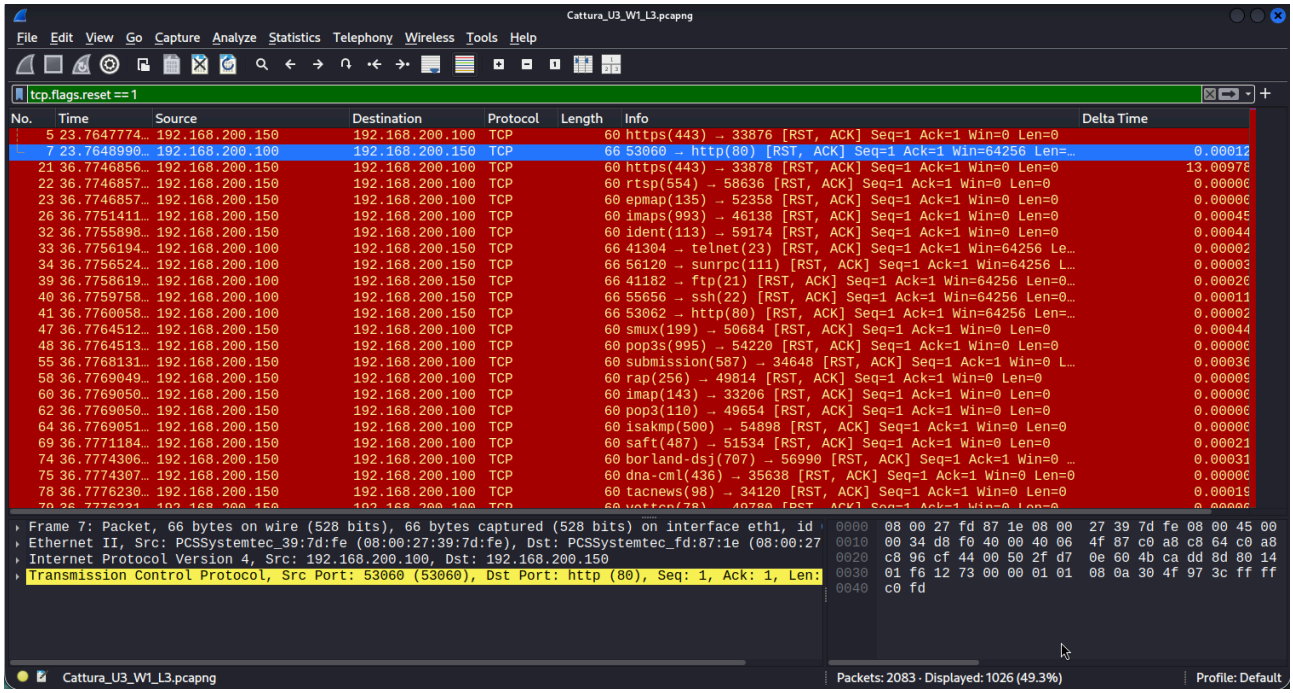


Figure 2 – TCP RST responses from target host



These figures clearly illustrate the behavior of a TCP SYN scan, where multiple connection attempts are made without completing the TCP handshake, followed by RST responses from the target.

3. Indicators of Compromise (IOCs)

The following IOCs were identified:
- Large number of SYN packets without completing TCP handshake
- Repeated RST responses from the target host
- Sequential port probing behavior
- Automated and repetitive traffic pattern consistent with scanning tools

4. Attack Hypothesis

The observed behavior is consistent with a TCP SYN Scan (half-open scan).

In this scenario:
- Source IP (192.168.200.100) is likely the attacker
- Destination IP (192.168.200.150) is the target

The purpose of this activity is to identify open ports and available services, which could later be exploited.

5. Mitigation Measures

To reduce the impact of this activity, the following actions are recommended:
- Block the suspicious IP address at firewall level
- Deploy IDS/IPS solutions (e.g., Snort)
- Implement rate limiting on incoming TCP connections
- Monitor network traffic continuously for anomalies
- Apply hardening techniques to exposed services

6. Conclusion

The analysis revealed active network scanning activity within the environment, which is commonly associated with the reconnaissance phase of an attack. Proper monitoring and defensive measures are essential to prevent further escalation.

CSIRT Italy Phishing Campaign Analysis

1. What is CSIRT Italy (ACN)?

CSIRT Italy (Computer Security Incident Response Team) is the national cybersecurity response team of Italy, operating under the National Cybersecurity Agency (ACN). It is responsible for monitoring, analyzing, and responding to cybersecurity threats and incidents affecting national infrastructure, organizations, and citizens.

2. What are its responsibilities?

The main responsibilities of CSIRT Italy include:
- Monitoring and analyzing cyber threats and vulnerabilities
- Issuing alerts and advisories on ongoing cyber campaigns
- Coordinating incident response activities
- Supporting organizations in managing and mitigating cyber incidents
- Sharing threat intelligence and Indicators of Compromise (IOCs)
- Enhancing national cybersecurity resilience

3. Analysis of the Alert (Trenitalia Phishing Campaign)

The analyzed alert describes a phishing campaign impersonating Trenitalia. The attackers use emails that appear legitimate, offering users the opportunity to obtain an annual pass by participating in a fake survey.

Attack flow:
- The victim receives a phishing email
- The email invites the user to complete a survey
- After completing the survey, the victim is informed of a fake reward
- The victim is redirected to a malicious page
- Sensitive data is requested (personal and payment information)

Impact:
- Theft of personal data
- Financial fraud (credit card data collection)
- Potential identity theft

4. How to protect an organization from this phishing campaign

To mitigate the risks associated with this phishing campaign, the following measures should be implemented:

Technical controls:
- Email filtering and anti-phishing solutions
- Domain and URL filtering
- Implementation of SPF, DKIM, and DMARC
- Deployment of IDS/IPS systems

Organizational measures:
- Regular security awareness training for employees
- Phishing simulation exercises
- Clear reporting procedures for suspicious emails

User behavior:

- Do not click on suspicious links
- Verify sender email domains
- Avoid entering sensitive data on untrusted websites
- Always verify offers that seem too good to be true

5. Conclusion

The phishing campaign analyzed demonstrates how attackers exploit brand trust to deceive users. Organizations must combine technical defenses, user awareness, and continuous monitoring to effectively mitigate such threats.